

Oslo, Norway
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Dear Reviewers,

We thank the reviewers for their careful reviews overall and for the constructive final comments. We believe that their feedback has significantly improved our manuscript.

Our detailed responses to the reviews follow below and in the subsequent pages. The reviewers' comments are included in italics, while our responses follow in roman text. The changes are included verbatim in this response letter and a version of the manuscript with changes marked (in the main text) is also attached for completeness..

Reviewer #1

Reviewer #1 (Remarks to the Author):

I would like to thank the authors for taking my feedback into account. The authors have made substantial strides towards addressing my main concern, namely the unclear placement of the manuscript between a methods paper and a biomedical paper. One easily fixed issue related to it is the wording of the abstract: "In this work, we combine mixed-dimensional mathematical modelling, multi-modal magnetic resonance images, and high performance computing to present a computational framework that predicts the spatiotemporal spreading of a solute across the subarachnoid space, ...". The term 'predicts' implies correct reproduction of the actual spatiotemporal map, but this has not been demonstrated in the manuscript. A wording along the lines of "a computational framework for calculating" should be used instead.

We thank the reviewer for their constructive feedback throughout.

In the abstract, we have replaced the phrase "that predicts" by "for calculating" as suggested by the reviewer. The revised sentence reads as:

"In this work, we combine mixed-dimensional mathematical modelling, multi-modal magnetic resonance images, and high performance computing to present a computational framework **for calculating** the spatiotemporal spreading of a solute across the subarachnoid space, ventricular system, and brain parenchyma, including surface perivascular spaces."

Reviewer #3

Reviewer #3 (Remarks to the Author):

This version of the manuscript includes major changes (both in the modeling choices and in the MS content) that considerably improve clarity. With regard to content, the perspective adopted focuses on the introduction of a new computational framework and better describes the limitations (not only due to mathematical and computational aspects but also due to the lack of biophysical understanding of the flow and transport mechanisms within the parenchyma). With regard to modelling choices, the treatment of interface conditions between PVS leaf nodes and parenchyma, which I argued was inconsistent in my previous reviews (see Major Concern A in review on Revision 1), has been fully revisited.

We thank the reviewer for their careful and highly constructive feedback throughout the revisions.

In my understanding, in this new version, the flow at these PVS leaf nodes is fully set up by the peristalsis model (lines 391-408) while all tracer mass reaching these points (by advection and diffusion) is exchanged with a 0D compartment that represents the downstream part of the PVS network, then exchanging diffusively with the 3D parenchyma. Notably, the contribution of pressure gradients induced by CSF production is now fully disregarded, thus enabling the “pressure continuation” approach from the previous versions to be abandoned. In other words, for the resolution of the flow problem, the flow, only induced by peristalsis, is fully “absorbed” in the 0D compartments, while for the resolution of the transport problem, the whole tracer mass advected toward the PVS leaf nodes is “free” to exchange with the 0D compartments (only limited by the diffusive permeability coefficient ξ_{EF}).

I understand that, given the current uncertainty on the flow pathways in the CNS, the authors have to make choices, but I would like them to underlie this more strongly in the Introduction and Discussion: the underlying flow field is strongly controlling mass transport, so that the idea of using a decoupled approach for the resolution of CSF flow, which is now clearly stated, is a fundamental limitation of this work. For example, assuming an advective contribution in the parenchyma (glymphatic hypothesis) versus a flow further downstream along PVS spaces ultimately reaching large CSF spaces (at the surface or ventricular level) would possibly yield very different results for the tracer concentration dynamics.

We interpret the reviewer’s comment as a suggestion to clearly point out, both in the Introduction and in the Discussion, our decoupled approach to CSF and PVS flow modelling and that this is a fundamental limitation of the work. In response, we have made the following additions to the revised manuscript.

First, we have added a sentence in the final paragraph of the Introduction as follows:

... we account for heterogeneous diffusion, dispersive mixing induced by cardiac and respiratory pulsatility in the CSF spaces, convective fluid flow driven by CSF production in the CSF spaces and peristaltic pumping in the PVSs, as well as solute

exchange and efflux across semi-permeable membranes. A fully coupled model of CSF, ISF, and PVS flow is however not within the scope of the current work. We demonstrate the applicability of this computational framework via a series of model scenarios ...

Second, we have added the following sentence at the end of the first paragraph of the Discussion:

In our simulations, the PVSs thus function as highways facilitating rapid transport, even in the absence of tight structural barriers. Note however that our PVS flow models are not coupled to flow in the SAS or within the brain parenchyma, a limitation of this work that is further discussed below.

I would like them to discuss more assertively

1/ the extent by which their modeling choices align with the different biophysical hypotheses that can be found in the literature, as suggested by their answer ("However, because our network end nodes are artificial truncations, enforcing a local mass balance at these specific coordinates would require us to introduce localized, artificial sinks/sources in the parenchyma and SAS. Doing so would effectively hard-code a specific (and highly debated) fluid pathway into the model at arbitrary geometric locations");

This is a very constructive suggestion and one that is further specified in the reviewer's Specific comment A below. Our response is provided in connection with that comment.

2/ which fundamental modelling challenges still need to be overcome in order to achieve a fully coupled model, such challenges being suggested in their answer ("The significant mathematical, numerical, and computational complexity of a fully coupled CSF/ISF/PVS flow model contributes to this consideration")

This is a very good suggestion, and we appreciate the opportunity to clarify. Concretely, we have added a sentence in the last paragraph of the Discussion, reading as follows:

Advancing this platform to a fully coupled flow framework requires a stable discretization of a 3D-1D Stokes-Darcy flow model that integrates vascular pulsations into a net flow field, alongside resolving open biophysical questions regarding PVS-tissue and PVS-SAS fluid exchange, and the boundary treatment of the artificially truncated network ends.

Specific comments:

A. Coming back to the author's comment: "However, because our network end nodes are artificial truncations, enforcing a local mass balance at these specific coordinates would require us to introduce localized, artificial sinks/sources in the parenchyma and SAS. Doing so would effectively hard-code a specific (and highly debated) fluid pathway into the model at arbitrary geometric locations". I would like them to be very clear on the hard-core biophysical mechanisms assumed by the current model. I don't think the new approach is fundamentally changing these, as suggested by the new Results Section in the Supplementary Information file

comparing this new approach with a modified version of the previous one only keeping peristalsis (S2.1 Effect of 0D-compartments at arterial leaf nodes and Figures S7 and S8). As underlined by the authors, the dynamics is not fundamentally changed (when looking at averaged quantities), which suggests both model account for similar mechanisms.

The reviewer asks for a clear description of the biophysical mechanisms assumed by the current model – specifically in connection with the fluid exchange between the PVS network ends and the parenchyma/SAS. In response, we have made the following two modifications to the first section of Results. (This placement was also suggested by the reviewer, cf. the comment C below). First, we have split the description of the 0D domains, now reading as:

... networks of topologically one-dimensional (1D) curves embedded therein (Figure 1B-F). We also associate the arterial leaf nodes of the PVS networks with zero-dimensional (0D) compartments representing the unresolved downstream perivascular pathways (see Methods).

Second, we have added the following description of the biophysical mechanisms assumed by the current model (citations replaced by [...]):

... convection via net fluid flow induced by peristaltic pumping[...]. At the arterial PVS leaf nodes, we assume that the solute can distribute evenly into (or exit from) a local region of the surrounding 3D domain, at a rate governed by the difference between the average concentration in said region and at the leaf node and a diffusive conductance, together with the convective contribution from the periarterial net flow.

B. However, as also pointed out by the authors, the details of the localization and size of ball representing these 0D compartments matter. To better illustrate this, I suggest to add a panel highlighting the relative difference between panels G and H in Figure S8

In response to the reviewer's constructive suggestion, we have moved the previous panels G and H into a new Figure S9 as panels A and B, respectively, and added a panel highlighting the relative difference in Panel C in said Figure.

C. The sentence added to clarify the key modelling contribution of the MS "We remark that while we present a coupled framework for solute transport in the CSF spaces, brain and surface PVS networks, a comprehensive model of CSF and ISF flow is not within the scope of this work. In particular, we here demonstrate using separate; i.e., not coupled, flow models in the CSF and PVS" (lines 143) should be integrated within the first Section of Results "A computational framework for intracranial solute transport".

This is a good suggestion. In response, we have made minor corrections to said sentence (replacing "brain" by "brain parenchyma", "CSF" by "CSF spaces"), and moved it from the second to the first Section of Results. In the revised manuscript, the sentence now reads as

... Mathematically, the solute transport model is represented by a mixed-dimensional system of coupled time-dependent partial differential equations[...], which we solve

numerically with high accuracy using a mass-conserving finite element scheme. We remark that while we present a coupled framework for solute transport in the CSF spaces, brain parenchyma, and surface PVS networks, a comprehensive model of CSF and ISF flow is not within the scope of this work. In particular, we here demonstrate using separate; i.e., not coupled, flow models in the CSF spaces and PVS. The complete computational framework is implemented in the FEniCS finite element software[...], and all associated software is all openly available[...].

Additional text could be added at this same location to address my comment in A.

Added as suggested: see our response to the comment in A above.

D. I also encourage the authors to be more assertive in the Discussion, in line with their answer to my comment: "Our rationale for this is, in part, the lack of consensus and significant debate surrounding the precise anatomical and functional nature of fluid exchange between the PVS, the SAS and interstitium, e.g. within surface and penetrating perivascular spaces, and between the PVS and surrounding CSF spaces and/or neuropil", by rewording the text (Line 297) : "based on the currently incomplete knowledge" to explicitly mention a lack of consensus in the field.

We have adopted the reviewer's suggestion and reworded the text. This sentence now reads as:

Our choice – to consider decoupled flow models – is based on the **current lack of consensus** on the underlying physiology, the mathematical and computational complexity associated with coupled mixed-dimensional fluid flow models especially the auxiliary modelling choices to be made regarding interaction at the network end nodes, and our lack of more complete vascular network data.

E. Description of PVS outer surface: please clearly state that the modelling choice represent a full physical compartmentalization with regard to flow, while enabling diffusive exchanges throughout this barrier (see in particular lines 220 and 276) (in other word, closer to reports from Weller and al. than Bakker and al.)

Action: The text has been revised as suggested by the reviewer. Concretely, the sentence in line 220 of the original manuscript, now reads as:

Our models of CSF flow in perivascular networks, based on **a full physical compartmentalization of the PVS in combination with** first principles and asymptotic analysis, predict net flow of such magnitudes and both antegrade and retrograde flow within the periarterial network.

In addition, the sentence in line 276, has been revised as follows.

In our simulations, the PVSs act as rapid transport pathways even with only a partial barrier to diffusion between the PVS and surrounding CSF, **though under the**

assumption that the PVS are fully physically compartmentalized with regard to flow. Indeed, we emphasize that we report ...

Other comments:

- Figure 1 Legend please add definition of acronyms used in the legend of panels B and C
- Figure 2 Legend please add definition of acronyms used in the legend of panels D and E (in particular PVS cont).

We added the definitions of the acronyms in the legends as suggested.

- Line 220 *"In our simulations, the PVSs thus function as highways facilitating rapid transport, even in the absence of tight structural barriers". I still find this formulation ambiguous. In the current framework, there is always a tight structural barrier channeling peristaltic flow within the PVS, as more clearly stated on line 276.*

We clarified the sentence by specifying

In our simulations, the PVSs thus function as highways facilitating rapid transport, even in the absence of tight structural barriers [to solute exchange](#).

- Line 222 : *tracer spread of the brain => tracer spread within the brain ?*

Corrected as suggested.

- Line 284-285: *"The high dispersion scenario considered here can be viewed as a demonstration of additional pulsatility induced by e.g. neural waves or other respiration patterns depending on activity level." => "The high dispersion scenario considered here can be viewed as representative of additional pulsatility induced by e.g. neural waves or other respiration patterns depending on activity level."*

Corrected as suggested.

- Line 330. *Is "white matter" needed here ?*

We removed "white matter" as suggested.

- SI line 119 : *suppress "Here".*

We removed "Here" as suggested.

- SI line 253 : *"To evaluate whether a simpler boundary condition is" I am not sure it is much simpler. Please replace by alternate or equivalent term.*

We rephrased to "alternative formulation" as suggested.